

Case Study: Network Architecture Powered by IBM

Estimated time needed: 10 minutes

As part of The Co-op Group, a diverse family of businesses spanning retail, insurance, and more, the Co-operative Bank bank's IT systems had become tightly intertwined with the rest of the group's network. When the bank planned to become a separate entity, this shared strategy had to change. The bank had to reduce its reliance on the group's IT services and take control of its own systems. This called for an extensive transformation program.

The bank was under heightened regulatory scrutiny due to recent high-profile IT issues within the UK banking sector. Every aspect of the program's planning, preparation, and execution must be meticulously managed.

While The Co-operative Bank's IT team had the expertise to manage their applications and infrastructure, their focus was primarily on day-to-day banking operations. Building, maintaining, and documenting systems that could handle the pressures of daily operations had been their core concern. However, the transformation program presented an additional challenge. Separating the bank's systems and networks from The Co-op Group's infrastructure meant migrating to an entirely new environment: new data centers, new servers, new networking, and a new operational model. The magnitude of these changes raised the risk of unforeseen behavior and unpredictable consequences. With the financial data of hundreds of thousands of customers on the line, the bank couldn't afford to leave anything to chance.

"Whilst the IT separation project was crucial for the bank's future, our commitment to our customers always comes first," emphasizes Chris Davis, Chief Operating Officer of The Co-operative Bank. "We needed absolute confidence that we could execute this transformation without any impact on the customer experience. That's why we chose to partner with IBM – their unparalleled expertise in large-scale banking migration projects gave us the certainty we needed."

Large migration effort

Co-operative Bank partnered with IBM to pursue a solution.

Planning and executing the transformation

To design a blueprint and roadmap for the transformation program, The Co-operative Bank collaborated closely with executive architects from IBM® Consulting. Most of the work was divided into two key phases: the Enterprise Services program and the Separation program.

Enterprise Services program

The Enterprise Services program focused on creating two cutting-edge data centers with state-of-the-art business continuity and disaster recovery capabilities. It also involved migrating the bank's priority business services, including core banking applications and treasury and finance systems.

The program was structured in six "waves," each encompassing a set of business-critical applications seamlessly moving to the new infrastructure. Cutovers were carefully scheduled over weekends to minimize the bank's and its customers' impact. The planning and execution of these waves utilized advanced methodologies that IBM has successfully employed in numerous flagship financial services migration projects. These methodologies are well-documented in the IBM Cloud® Innovate repository, now part of the IBM Garage™ Method for Cloud.

The waves were designed to reduce risk by creating a protective "bubble" around the migrated applications. In essence, each wave began by establishing the necessary data center infrastructure and deploying a copy of the applications and data onto it. The team then rigorously tested the hardware, operating systems, applications, data, and batch processes to ensure seamless functionality in the target environment. Once everything within the bubble passed its tests, the team proceeded with one or more "dress rehearsals" to instill confidence that the cutover event could be executed successfully within the allotted time. Finally, the team performed the cutover itself.

During its peak, the Enterprise Services program was supported by a dedicated team of 700 people, including over 300 IBM colleagues. These resources were strategically aligned with specific workstreams within each wave, such as infrastructure, application migration, and testing. The consistent leadership across all six waves ensured continuity. It enabled later waves to benefit from the knowledge gained during earlier migrations.

Separation program

The Separation program consisted of diverse projects aimed at facilitating the bank's smooth transition and complete technology separation from The Co-op Group's infrastructure. These projects were organized into independent "towers," each managed by a dedicated team. Let's take a closer look at these towers:

- **Application migration:** This tower focused on migrating or decommissioning over 100 applications and technical tools for monitoring and data transfer. These systems often had intricate dependencies. For instance, one service interacted with over 100 data feeds, each provided by a different third party.
- **Application modification:** The emphasis was on modernizing several larger applications before relocating to the new data centers.
- **Data warehousing:** This tower focused on upgrading the bank's extensive serial attached SCSI (SAS) analytics environment and consolidating various SAS systems into a new, bank-exclusive single instance.
- **Legacy:** The focus of this tower was to assume responsibility for the infrastructure hosting and network support of a select few non-strategic services running in the existing bank-owned data centers, which are scheduled for replacement soon.
- **Security:** This tower aimed to phase out a range of security services the bank had provided to the rest of the group.
- **External connections:** Migration of several third-party virtual private network (VPN) connections to the target infrastructure fell under this tower's purview.
- **IT management services:** This tower focused on services shared between the bank and the group. For example, separating the user directory into distinct instances for each organization.
- **Network:** The network tower was responsible for ensuring the separation of networks between the bank and group networks. The network team avoided a "big bang" cutover to minimize risk. Instead, it transitioned the bank's landscape through seven intermediate states to prepare for the final separation.

The bank aimed to ensure a seamless and successful technology separation by undertaking these initiatives, setting the stage for future growth and innovation.

Achieving a zero-disruption migration

Both phases were executed flawlessly, and the bank hails the transformation as an unequivocal triumph. In the Enterprise Services program, the cutovers went without a hitch. Take, for instance, the core banking wave, the most colossal and intricate of the six, which wrapped up over four hours ahead of schedule, with not a single severity-one issue post-cutover weekend. The very next day turned out to be the "quietest Monday we'd had in months," as one member of the bank's IT leadership team put it.

Likewise, after the final technology separation that marked the culmination of the Separation program, the bank encountered no severity-one or severity-two issues—an extraordinary feat given the magnitude and scale of the transformation it undertook.

During the project, Mark Record, the IT & Change Director at The Co-operative Bank, worked hand in hand with the IBM team throughout the Separation program. Reflecting on the success, he remarks, "Above all, the project thrived because we assembled the right people on the team. Our bank team brought years of invaluable knowledge about our applications and systems. At the same time, IBM contributed an unparalleled level of migration expertise. With the perfect blend of resources, we had complete confidence in the outcome—and even though the project faced intense scrutiny, it was a matter of trusting the process and 'letting the pilots fly the plane.'"

He adds, "IBM's migration methodologies are in a league of their own, but even the finest methodology can't resolve every challenge in a project of this magnitude. When faced with an insurmountable problem, you need individuals who can swiftly delve deep into a system, and that's precisely what IBM brought to the table."

For instance, during the Enterprise Services program, the team encountered an application that functioned flawlessly in the source environment but failed to run correctly during bubble testing in an ostensibly identical target environment. An assigned IBM Distinguished Engineer meticulously reviewed every application and middleware configuration setting alongside the relevant teams and collaborated with IBM product subject-matter experts and bank subject-matter experts to identify the problem: a deeply buried issue within the operating system. Once diagnosed, it was effortlessly resolved. As one stakeholder at the bank aptly remarked at the time, "No matter the challenge they face, this team will always find a solution."

Overall, the team triumphed in migrating over 250 applications operating on more than 800 servers across both transformation programs. Across the myriad technical implementations, they achieved a 100% success rate. They concluded the final technology separation with virtually no impact on the business and its customers.

Consequently, the bank now boasts two world-class data centers and an advanced disaster recovery capability, providing a resilient platform for day-to-day operations and a solid foundation for future digital transformation. The organization has reduced its technical debt through application modernization, enabling greater agility in developing digital services and launching new products to market.

Andrew Bester, the bank's Chief Executive Officer, concludes, "The transformation program we've accomplished with IBM has empowered us to establish an independent architecture that gives us full control of our IT strategy. As we continue to forge ahead with new digital services and embrace the cloud, we now possess the technological groundwork required to deliver faster, smarter, and more convenient banking experiences to hundreds of thousands of retail and business customers across the UK."

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